

Strict-past Gaussian causalization and resolvent signed-charge reduction

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claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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Purpose and scope

This note proves the finite-dimensional strict-past identity that prevents a predictable shell source from charging the Gaussian past energy at every scale. It closes the scoped child

A13-CLASSII-STRICT-PAST-RESOLVENT-SIGNED-CHARGE-REDUCTION.

The result converts the controlled-shell lower bound into one signed charge: the positive predictor square is kept together with the divergence of the innovation. There is one control-energy payment, not one payment per shell.

The theorem is exact for a finite ordered Gaussian filtration. It does not bound the resulting production charge uniformly in the cutoff. In particular, it does not by itself control the centered derivative-square tensor Q , the rational current, the translated-frame square, or the Cartan curvature term. The current selected gate

A13-CLASSII-STRICT-PAST-SIGNED-Q-CURRENT-FORM-BOUND

and its parent one-use/Nelson gate remain open.

1. Ordered Gaussian convention

Let $\xi = (\xi_1, \dots, \xi_N)$ be standard Gaussian and let $\mathcal{F}_j = \sigma(\xi_1, \dots, \xi_j)$. A map $a : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is strictly past-adapted when a_j is measurable with respect to $\mathcal{F}_{j-1}$. For smooth a define the Gaussian divergence

$$\delta a(z) = \sum_{j=1}^N \{z_j a_j(z) - \partial_j a_j(z)\}. \quad (1.1)$$

Let h be strictly past-adapted and put $Z = \xi + h(\xi)$. Let $b = b^< + r$, where $b^<$ is strictly past-adapted in its argument and r is the unstructured remainder. The ordering may encode all coordinates in one sharp shell before moving to the next shell; strict past then means that the present shell never enters its own predictor.

Strict triangularity implies

$$\partial_j b_j^< = 0, \quad \text{Tr}\{D b^<(Z) D h(\xi)\} = 0. \quad (1.2)$$

The second identity records the absence of same-shell feedback. It is not a smallness estimate.

2. Exact causalization lemma

Lemma 2.1 (strict-past Gaussian causalization). Assume that all expressions below are integrable. Then

$$\mathbb{E} \delta b^<(Z) = \mathbb{E} \langle h(\xi), b^<(Z) \rangle. \quad (2.1)$$

Proof. For each j , both $h_j(\xi)$ and $b_j^<(Z)$ are $\mathcal{F}_{j-1}$ -measurable. Indeed, the triangular recursion shows that $Z_1, \dots, Z_{j-1}$ depend only on $\xi_1, \dots, \xi_{j-1}$. Hence ξ_j is independent of $b_j^<(Z)$ and is centered. By (1.2),

$$\mathbb{E} \delta b^<(Z) = \sum_j \mathbb{E}[(\xi_j + h_j(\xi)) b_j^<(Z)] \quad (1)$$

$$= \sum_j \mathbb{E}[h_j(\xi) b_j^<(Z)], \quad (2)$$

which is (2.1). $\square$

The cancellation in (2.1) is conditional shell by shell, but the equality is an expectation identity. The integrand $\delta b^<(Z)$ cannot in general be replaced pointwise by $h \cdot b^<(Z)$.

3. One-use signed-charge reduction

Suppose the finite controlled interaction has the divergence form

$$2V(Z) = \delta b(Z) = \delta b^<(Z) + \delta r(Z). \quad (3.1)$$

For $\varepsilon > 0$ define the coupled signed charge

$$\mathcal{S}_\varepsilon(Z) = \frac{|b^<(Z)|^2}{16\varepsilon} - \frac{1}{2}\delta r(Z). \quad (3.2)$$

Theorem 3.1 (one-use reduction). Under the hypotheses of Lemma 2.1,

$$\mathbb{E}V(Z) = -\varepsilon\mathbb{E}|h|^2 - \mathbb{E}\mathcal{S}_\varepsilon(Z) + \varepsilon\mathbb{E}\left|h + \frac{b^<(Z)}{4\varepsilon}\right|^2, \quad (3.3)$$

and therefore

$$\mathbb{E}V(Z) \geq -\varepsilon\mathbb{E}|h|^2 - \mathbb{E}\mathcal{S}_\varepsilon(Z). \quad (3.4)$$

Proof. Equations (3.1) and (2.1) give

$$\mathbb{E}V(Z) = \frac{1}{2}\mathbb{E}\langle h, b^<(Z) \rangle + \frac{1}{2}\mathbb{E}\delta r(Z). \quad (3.5)$$

Complete exactly one square:

$$\frac{1}{2}\langle h, b^< \rangle = -\varepsilon|h|^2 - \frac{|b^<|^2}{16\varepsilon} + \varepsilon\left|h + \frac{b^<}{4\varepsilon}\right|^2. \quad (3.6)$$

Substitution yields (3.3), and deletion of only the final nonnegative expectation yields (3.4). $\square$

The source-square and the innovation divergence in (3.2) remain signed and coupled. Applying absolute values or Young inequalities to their individual shell constituents would undo the theorem: it would reintroduce repeated past-energy costs and discard the cancellation still needed from the Q /current/Cartan structure.

4. Positive affine block and resolvent completion

The production filtration contains more structure than Theorem 3.1 uses. Split the j th coordinate into a shell block and let $T_j = T_j(Z_{<j})$ be symmetric positive semidefinite and $\ell_j = \ell_j(Z_{<j})$ be strict-past measurable. Define

$$a_j(Z) = T_j Z_j + 2\ell_j. \quad (4.1)$$

Because T_j and ℓ_j do not depend on Z_j ,

$$\frac{1}{2}\delta_j a_j(Z) = \frac{1}{2}\{Z_j^T T_j Z_j - \text{Tr } T_j\} + \langle \ell_j, Z_j \rangle. \quad (4.2)$$

Conditioning on the strict past gives

$$\mathbb{E}_j \frac{1}{2}\delta_j a_j(\xi + h) = \frac{1}{2}h_j^T T_j h_j + \langle \ell_j, h_j \rangle. \quad (4.3)$$

Put $q = (2\varepsilon)^{-1}$ and $R_{q,j} = (I + qT_j)^{-1}$. The exact metric completion is

$$\frac{1}{2}h_j^T T_j h_j + \langle \ell_j, h_j \rangle = -\varepsilon|h_j|^2 - \frac{q}{2}\langle \ell_j, R_{q,j}\ell_j \rangle \quad (3)$$

$$+ \frac{1}{2}\left|(T_j + q^{-1}I)^{1/2}(h_j + qR_{q,j}\ell_j)\right|^2. \quad (4.4)$$

Thus the positive present-shell quadratic form is not discarded into the remainder. It rebates the raw source square by the resolvent $R_{q,j}$. For the production control allocation $\varepsilon = 0.45$,

$$q = \frac{1}{0.9} = \frac{10}{9}, \quad (4.5)$$

which is exactly the Nelson exponent isolated by the earlier Boue–Dupuis equivalence, not a new fitted constant.

5. Exact A11 production signed charge

Use the A11 notation

$$T_j = A_j^* A_j, \quad \ell_j = A_j^* r_j, \quad I_j = \frac{1}{2}\{\langle \xi_j, T_j \xi_j \rangle - \text{Tr } T_j\} + \langle \ell_j, \xi_j \rangle. \quad (5.1)$$

Equation (4.2) says $I_j = \delta_j a_j/2$. A11 also proves the exact finite-cutoff telescope

$$V_J^{\text{ren}} = \sum_{j \leq J} (I_j + \mathcal{C}_j), \quad V_0 = 0. \quad (5.2)$$

Let b_J be the A13 whitened production divergence field, so that $2V_J^{\text{ren}} = \delta b_J$, and set $r = b_J - a$. Equations (5.1)–(5.2) then give the pointwise algebraic identity

$$\frac{1}{2}\delta r = \sum_{j \leq J} \mathcal{C}_j. \quad (5.3)$$

Combining (4.4) over the shells with (5.3) yields the exact one-use identity

$$\mathbb{E} V_J^{\text{ren}}(X_J + h) = -\varepsilon \mathbb{E} \sum_j |h_j|^2 - \mathbb{E} \mathfrak{S}_{J,q}(X_J + h) \quad (4)$$

$$+ \frac{1}{2} \mathbb{E} \sum_j \left| (T_j + q^{-1}I)^{1/2}(h_j + qR_{q,j}\ell_j) \right|^2, \quad (5.4)$$

where

$$\mathfrak{S}_{J,q}(z) = \sum_{j \leq J} \left\{ \frac{q}{2} \langle \ell_j, R_{q,j} \ell_j \rangle - \mathcal{C}_j(z) \right\}. \quad (5.5)$$

Consequently the current child gate is reduced, without repeated past-energy payment, to

$$\mathbb{E} \mathfrak{S}_{J,10/9}(X_J + h) \leq \varepsilon_6 \mathbb{E} \|X_J + h\|_6^6 + C \quad (5.6)$$

uniformly for strict-past controls, with $\varepsilon_6 < \gamma/6$. The endpoint Cameron–Martin norm is bounded by the Boue–Dupuis control energy, so (5.6) implies the umbrella one-use estimate.

Formula (5.5) is the promised coupling. The first term is the A11 positive source-square with its exact resolvent rebate; the second is the complete coefficient increment. Its balanced A7 reconstruction contains the centered Q , rational current, translated-frame square, and Cartan curvature. The relative-phase factor-four effect has not vanished: it is retained in the dependence of $\mathcal{C}_j$ on the shifted current shell. Therefore a termwise or one-shell bound on the two summands is still forbidden. The remaining theorem is the global signed estimate (5.6), not a separated source bound and not a restatement of the Nelson moment.

6. Executable adversarial audit

The primary fixture uses tensor Gauss–Hermite quadrature on a nonlinear three-coordinate triangular shift, predictor, and unrestricted innovation. It checks (2.1), the trace cancellation, (3.3), the PSD affine identity (4.3), the resolvent completion (4.4), retention of the innovation divergence, and a point at which the corresponding pointwise completion gap is negative. The independent fixture does not import the primary program. It uses a different Hermite order and finite-difference Jacobians. It also breaks strict-past adaptation in two ways: a present-coordinate shift and a future-dependent predictor produce nonzero residuals. These are negative controls for the precise filtration hypothesis, not evidence for the infinite-dimensional production estimate.

Devil’s-advocate

1. **DISMISSED: the predictor cancellation is merely a fitted numerical pattern.** Lemma 2.1 is an exact conditional-expectation proof; two quadrature implementations are regression tests for conventions and signs.
2. **VALID WITH MITIGATION: (3.3) is a pointwise square identity.** Only (3.6) is pointwise after causalization. Replacing $\delta b^<(Z)$ by $h \cdot b^<(Z)$ is legal only after expectation. The primary negative control exhibits a negative pointwise gap.
3. **UPHELD AGAINST NONCAUSAL USE: an adapted shift may depend on the present or future shell.** Either dependence destroys the conditional centering step. The independent negative controls fail by macroscopic residuals.
4. **VALID WITH MITIGATION: the positive predictor square may be estimated separately from δr_J .** Formula (4.2) is deliberately a signed coupled charge. Separate absolute-value estimates can recreate the repeated energy loss that T-050 forbids.
5. **UPHELD AGAINST PROMOTION: causalization proves the production one-use bound.** It does not. The cutoff-uniform bound on (4.2), with the full Q /current/frame/Cartan algebra, is still open, so A13 remains T4.

Result footer

Result ID: A13-CLASSII-STRICT-PAST-RESOLVENT-SIGNED-CHARGE-REDUCTION

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

Precise statement: Strict-past causalization retains the PSD affine shell block and gives the exact q-resolvent charge. For A11, $\delta a/2=I_j$ and $\delta r/2=\sum C_j$, so one-use reduces to the global coupled charge $\sum_j[q_{\ell},R_{q\ell}]/2-C_j$.

Parameter range: Finite ordered standard Gaussian filtration; $\epsilon>0$. The audit pins $\epsilon=0.45$.

Dependencies: A11 true-increment divergence structure; A13 one-use/Nelson budget and strict sharp-cube order.

Tier before / after: T4 / T4 (no tier change).

Falsifiers: Present/future-shell dependence; nonzero causal identity residual; wrong q/resolvent factor or sign; failed primary, independent, PDF, hash, or release check.

No-overclaim: No cutoff-uniform production signed-charge bound, one-use theorem, Nelson moment, Gibbs measure, floor/regulator removal, infinite volume, T5-T7.

Next required action: Bound $S_-(J,10/9)$ with centered Q, rational current, translated-frame square, and Cartan curvature kept coupled under strict-past conditioning.